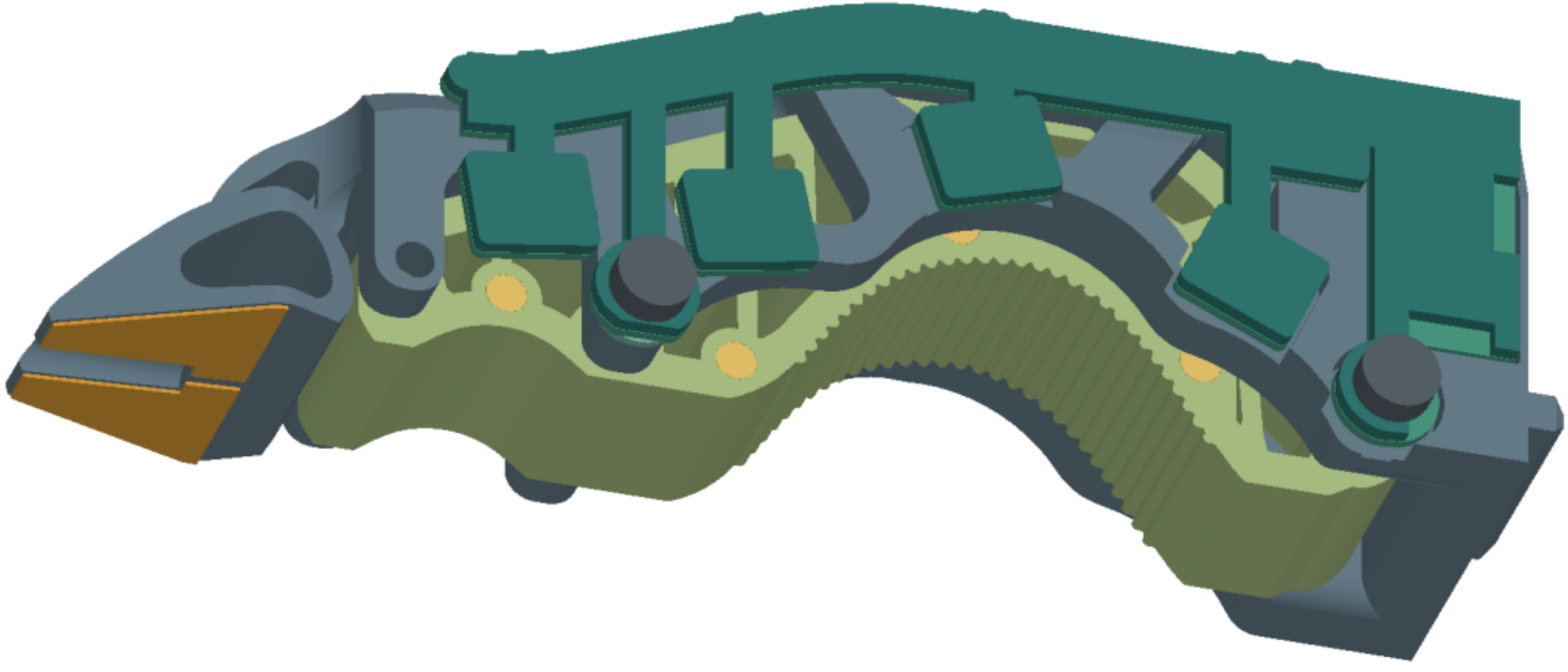
B3.1 · recessed board beside the compliant insert

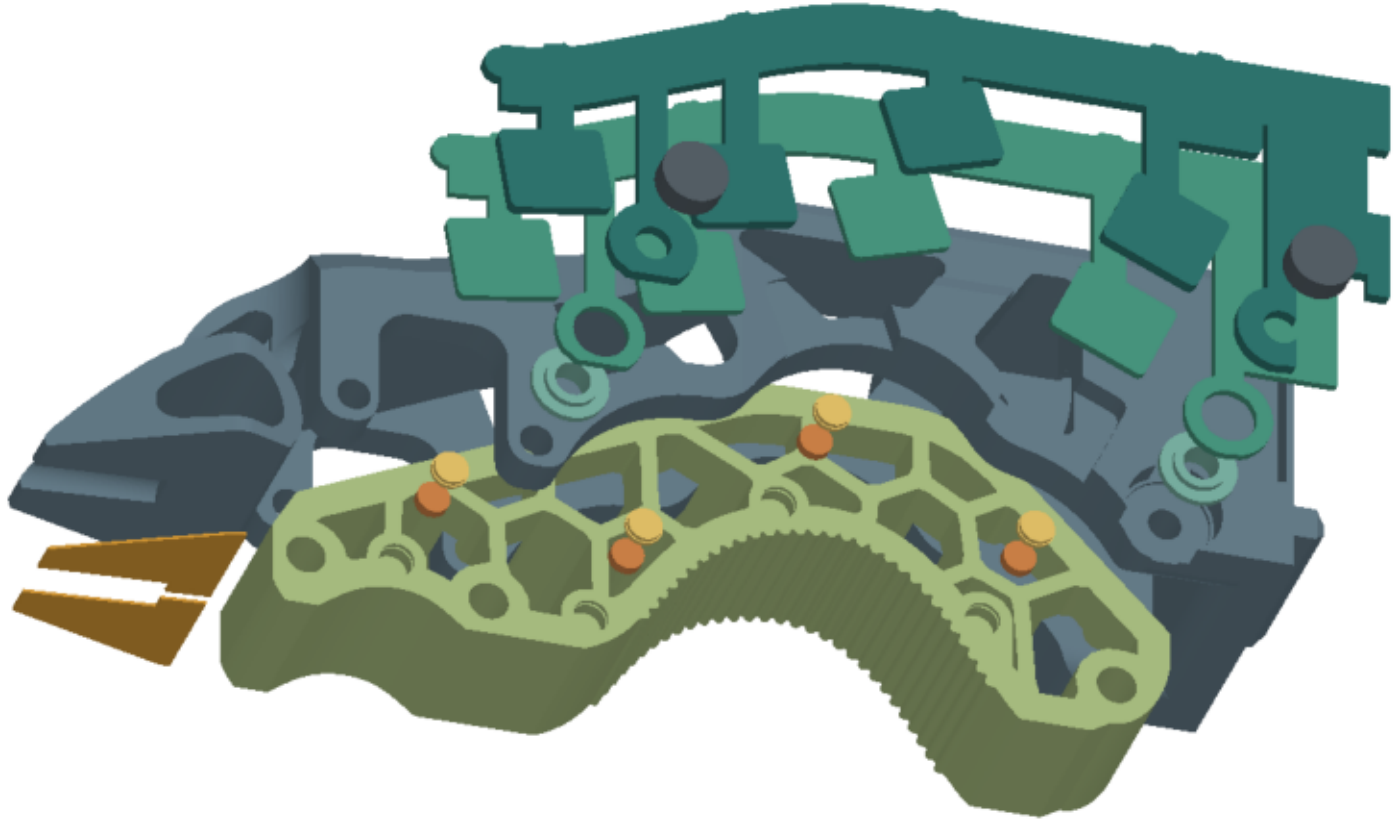


Revised frame · B1 TPU + four magnets · two N0 nail pads · B2 PCB · tapered spacers · backer

Actual CAD and package/head envelopes. Cable loops, coatings and seating film are omitted.

Experimental digital prototype · physical build, force calibration and 20 N qualification pending

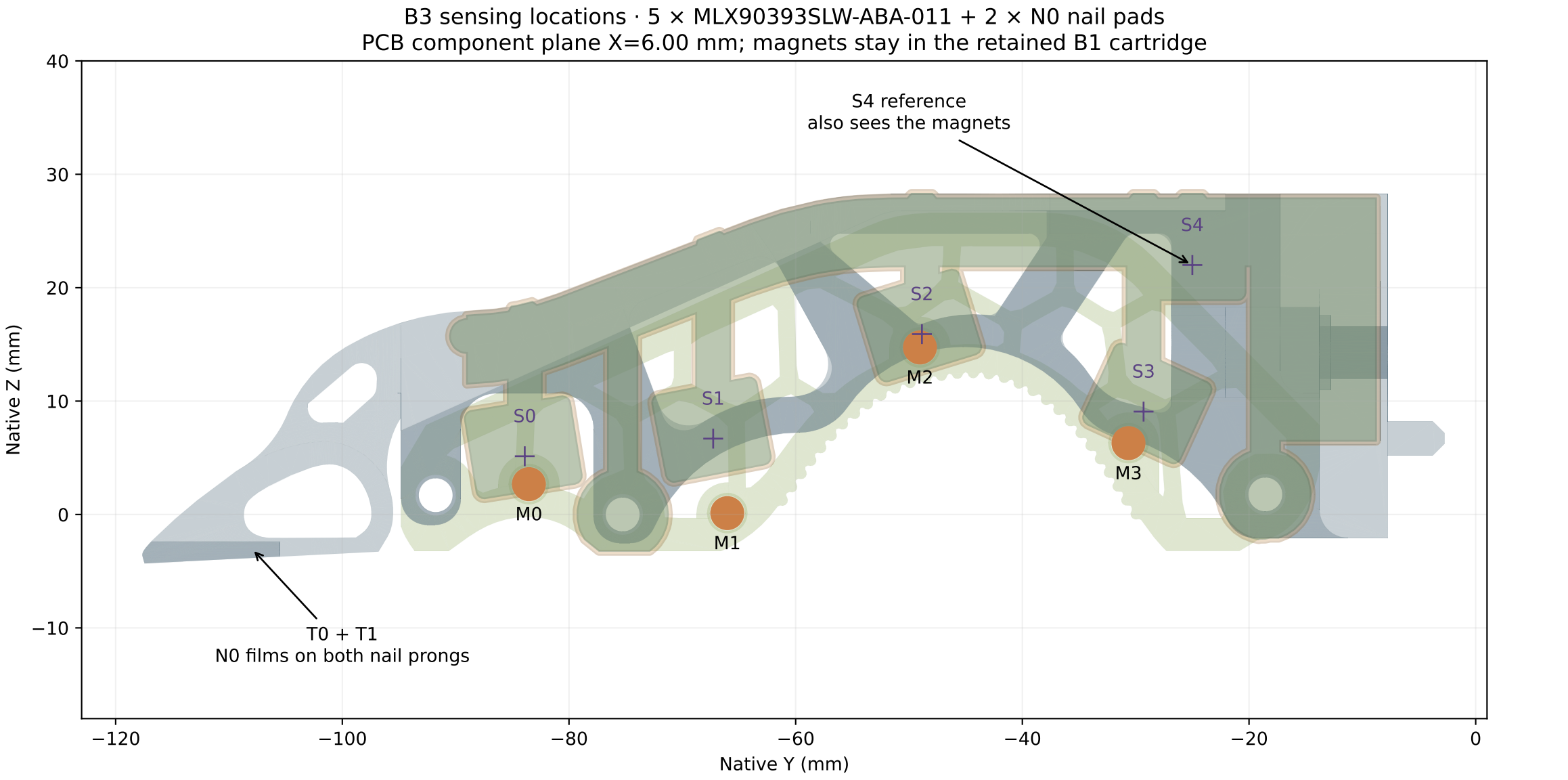
B3.1 · exploded recessed assembly

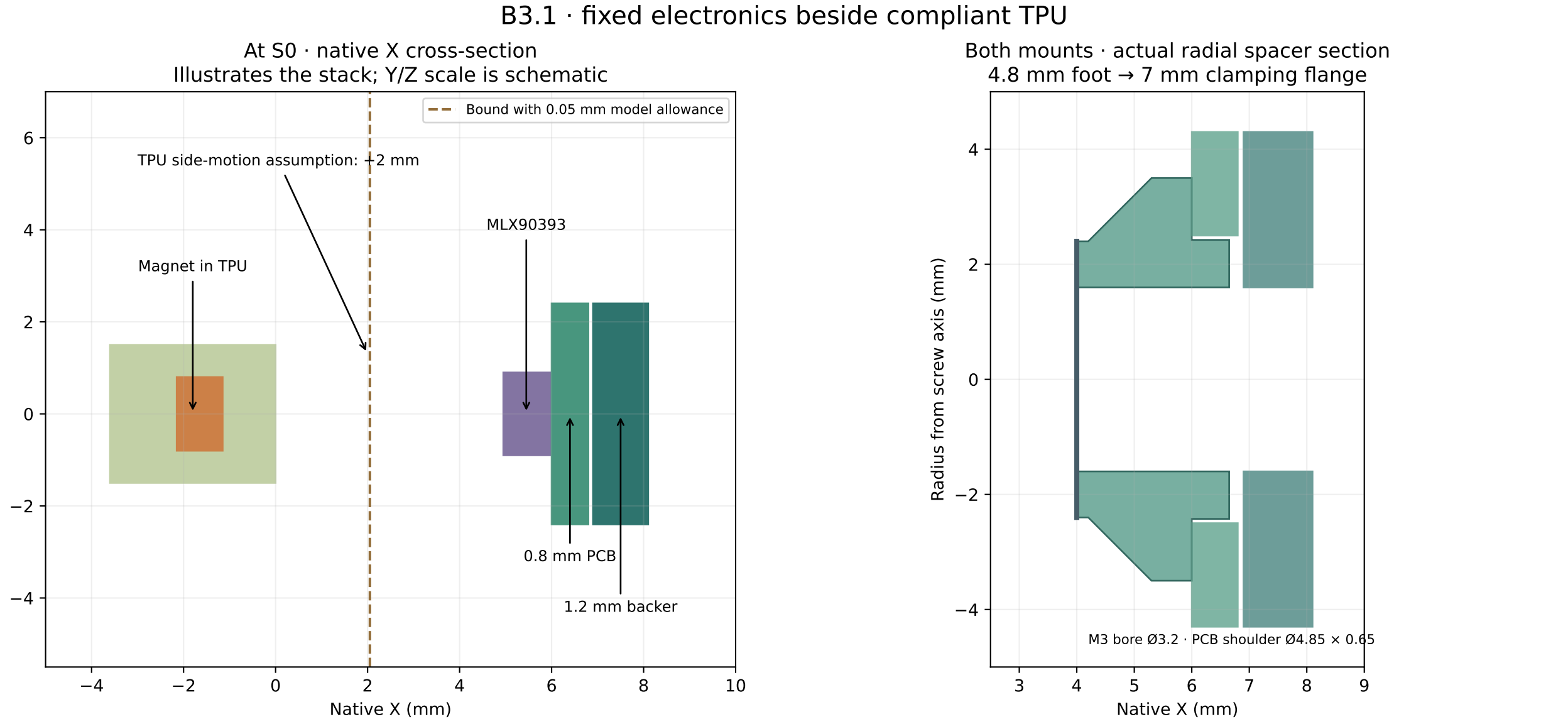


Revised frame · B1 TPU + four magnets · two N0 nail pads · B2 PCB · tapered spacers · backer

Actual CAD and package/head envelopes. Cable loops, coatings and seating film are omitted.

Experimental digital prototype · physical build, force calibration and 20 N qualification pending


B3 sensing locations · 5 × MLX90393SLW-ABA-011 + 2 × N0 nail pads
PCB component plane X=6.00 mm; magnets stay in the retained B1 cartridge
S4 reference
also sees the magnets
S0
S1
S2
S3
S4
M0
M1
M2
M3
T0 + T1
N0 films on both nail prongs
Native Z (mm)
Native Y (mm)
40
30
20
10
0
−10
−120
−100
−80
−60
−40
−20
0


B3.1 · fixed electronics beside compliant TPU
At S0 · native X cross-section
Illustrates the stack; Y/Z scale is schematic
Bound with 0.05 mm model allowance
TPU side-motion assumption: ±2 mm
MLX90393
Magnet in TPU
0.8 mm PCB
1.2 mm backer
Native X (mm)
Both mounts · actual radial spacer section
4.8 mm foot → 7 mm clamping flange
Radius from screw axis (mm)
M3 bore Ø3.2 · PCB shoulder Ø4.85 × 0.65
Native X (mm)

B3.1 review · millimeters · manufacture from STEP masters

Print one revised frame, one backer and two identical spacers.
PCB: retained B2.1 four-layer board; 0.8 mm nominal thickness.
Component face X=6.00; recess floor X=4.45.
Both spacer seats X=4.00; foot Ø4.80 × 0.20.
Taper length 1.10 to Ø7.00; full flange length 0.70.
PCB locating shoulder Ø4.85 × 0.65; central bore Ø3.20.
Board hole Ø5.00; backer inside X=6.90, outside X=8.10.
Seat film 0.10; keep leads out of the bearing annuli.

Added global width: 3.611 with reserved heads.
Head allowance Ø6.00 × 2.40; wires excluded; no added reach.
Select actual nonmagnetic M3 screw length from the measured stack.

Use four retained B1 3 × 1 mm N48 discs, north toward +X.
Five MLX90393SLW-ABA-011 ICs use GAIN_SEL=5, RES=1.
Nominal 25 C conversion: X/Y 0.501, Z 0.807 µT/count.
Two N0 nail cells retain 100 kΩ resistors at the pads.
Use B3's complete five-file firmware; do not reuse B2 scaling.

Check actual drive fit through full opening in both jaw positions.
The prescribed +2 mm side-motion bound is not a 20 N rating.
Print compliance, clamps, cable loops and strength are untested.
Protected source geometry does not establish unchanged stiffness.
No physical contact/force accuracy or specimen yield is reported.